2026

MARKET BASKET ANALYSIS

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PERSEVEX EDUCATION CONSULTANCY LLP

1. Introduction

Market Basket Analysis (MBA) is a data mining technique used to identify relationships and patterns between products that are frequently purchased together by customers. It plays a crucial role in the retail and e-commerce industries by helping businesses understand customer buying behavior and improve sales strategies through product recommendations and cross-selling techniques.

With the rapid growth of data in modern businesses, analyzing transaction data has become essential for making informed decisions. Market Basket Analysis uses association rule mining algorithms such as the Apriori algorithm to discover meaningful relationships between items. These relationships are measured using key metrics like Support, Confidence, and Lift, which indicate the frequency, probability, and strength of association between products.

In this project, Market Basket Analysis is performed using transactional data to identify strong product associations. The Apriori algorithm is implemented (using Python) to generate association rules, and the results are visualized using Microsoft Power BI. The interactive dashboard includes various visualizations such as scatter plots, bar charts, doughnut charts, and key influencer visuals, allowing users to explore insights effectively.

Additionally, advanced features like drill-through functionality are used to provide detailed product-level analysis, making the dashboard more interactive and user-friendly.

2. Objectives

The main objectives of this project are:

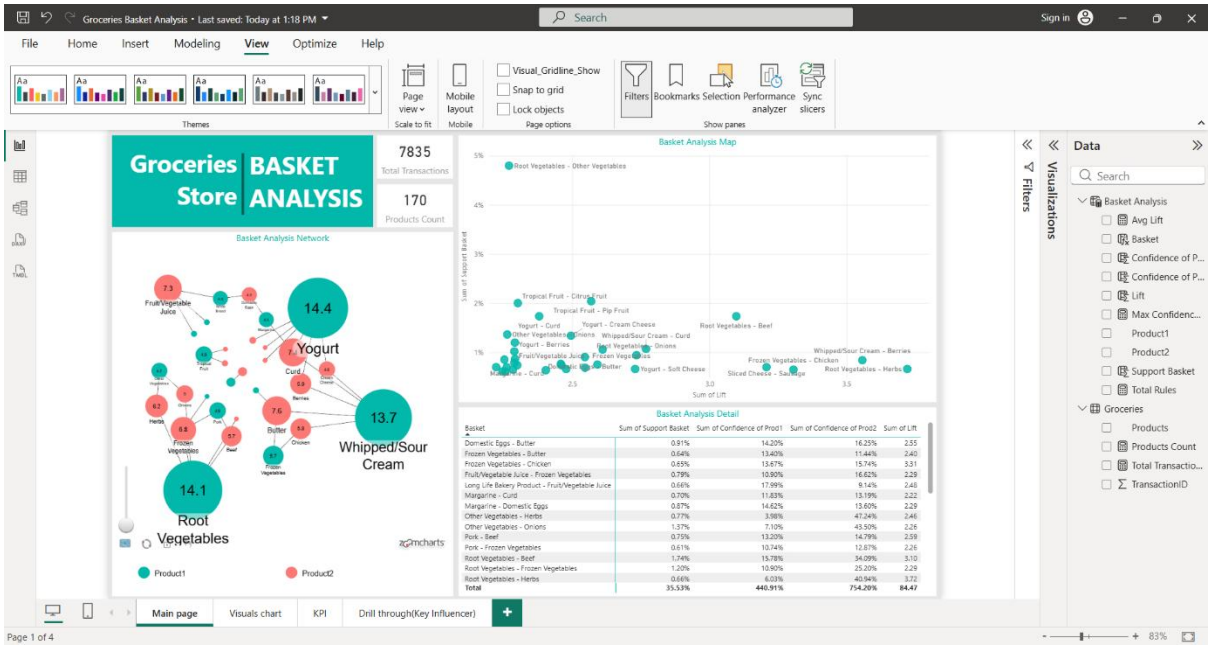
- To analyze customer transaction data and identify purchasing patterns using Market Basket Analysis.
- To apply the Apriori algorithm to generate association rules between products.
- To evaluate relationships between items using key metrics such as Support, Confidence, and Lift.
- To identify strong product combinations that can be used for cross-selling and recommendation strategies.
- To develop an interactive dashboard using Microsoft Power BI for better visualization and analysis of data.
- To use advanced visualization techniques such as scatter plots, doughnut charts, and key influencer visuals to gain deeper insights.
- To implement drill-through functionality for detailed product-level analysis.
- To support business decision-making by presenting insights in an easy-to-

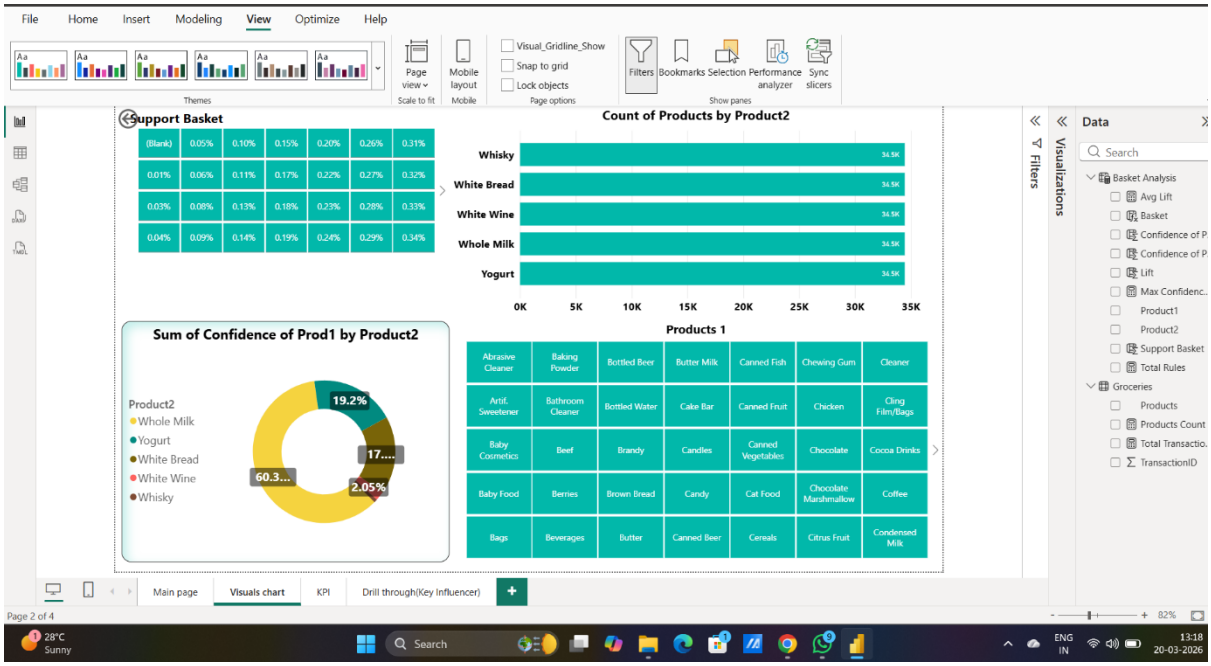
3. Uses

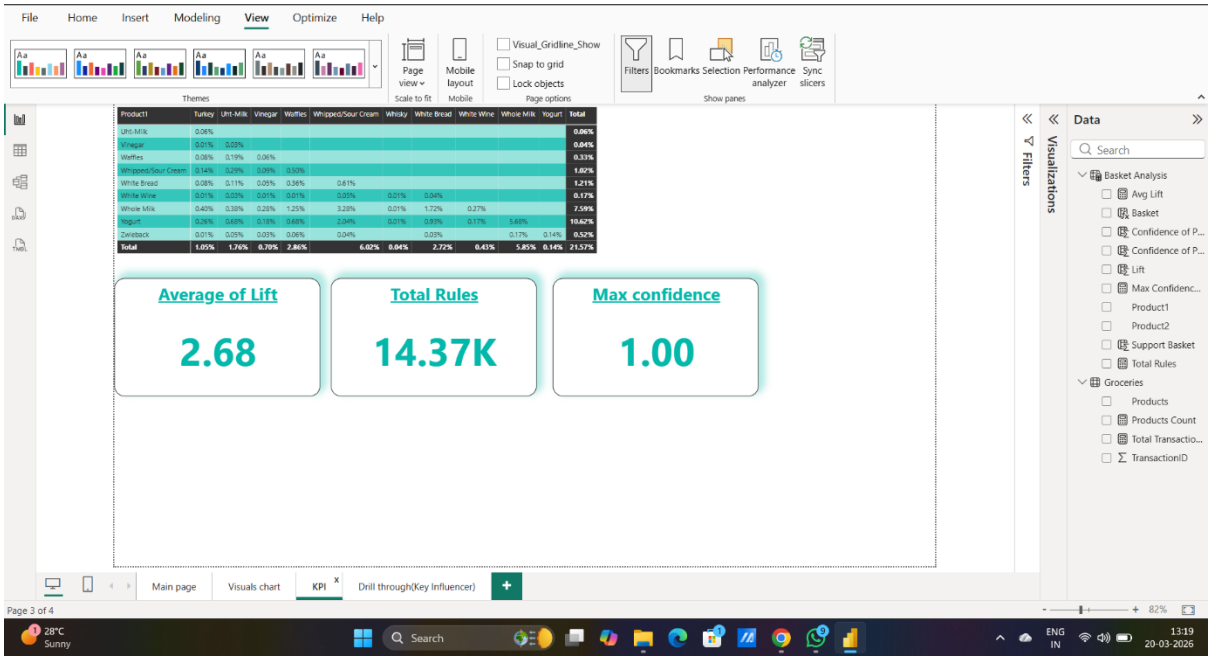
Market Basket Analysis has several practical applications in real-world business scenarios. The key uses of this project are:

- **Product Recommendation:**
  Helps businesses suggest related products to customers based on their purchase history (e.g., “Customers who bought this also bought...”).
- **Cross-Selling Strategies:**
  Identifies product combinations that are frequently purchased together, allowing companies to increase sales by promoting bundled products.
- **Store Layout Optimization:**
  Assists in arranging products in physical stores so that commonly bought items are placed closer to improve customer convenience and sales.
- **Targeted Marketing:**
  Enables personalized promotions and offers based on customer buying patterns.
- **Inventory Management:**
  Helps in maintaining stock levels by understanding demand for frequently associated products.
- **Business Decision-Making:**
  Provides insights that help managers make data-driven decisions using interactive dashboards built in Microsoft Power BI.
- **E-commerce Optimization:**
  Enhances user experience in online platforms by showing relevant product suggestions and increasing conversion rates.

4. Screen shots






File
Home
Insert
Modeling
View
Optimize
Help
Themes
Page view
Scale to fit
Mobile layout
Mobile
Visual_Gridline_Show
Snap to grid
Lock objects
Page options
Filters
Bookmarks
Selection
Performance analyzer
Sync slicers
Show panes
Product1 | Turkey | Unt-Milk | Vinegar | Waffles | Whipped/Sour Cream | Whisky | White Bread | White Wine | Whole Milk | Yogurt | Total
Unt-Milk | 0.06% | 0.06%
Vinegar | 0.01% | 0.03% | 0.04%
Waffles | 0.08% | 0.19% | 0.06% | 0.33%
Whipped/Sour Cream | 0.14% | 0.29% | 0.09% | 0.50% | 1.02%
White Bread | 0.08% | 0.11% | 0.05% | 0.36% | 0.61% | 1.21%
White Wine | 0.01% | 0.03% | 0.01% | 0.01% | 0.05% | 0.01% | 0.04% | 0.17%
Whole Milk | 0.40% | 0.38% | 0.28% | 1.25% | 3.28% | 0.01% | 1.72% | 0.27% | 7.59%
Yogurt | 0.26% | 0.68% | 0.18% | 0.68% | 2.04% | 0.01% | 0.93% | 0.17% | 5.68% | 10.62%
Zwieback | 0.01% | 0.05% | 0.03% | 0.06% | 0.04% | 0.03% | 0.17% | 0.14% | 0.52%
Total | 1.05% | 1.76% | 0.70% | 2.86% | 6.02% | 0.04% | 2.72% | 0.43% | 5.85% | 0.14% | 21.57%
Average of Lift
2.68
Total Rules
14.37K
Max confidence
1.00
Main page
Visuals chart
KPI
Drill through(Key Influencer)
Page 3 of 4
Data
Search
Basket Analysis
Avg Lift
Basket
Confidence of P...
Confidence of P...
Lift
Max Confidenc...
Product1
Product2
Support Basket
Total Rules
Groceries
Products
Products Count
Total Transactio...
TransactionID
Filters
Visualizations
28°C
Sunny
Search
ENG
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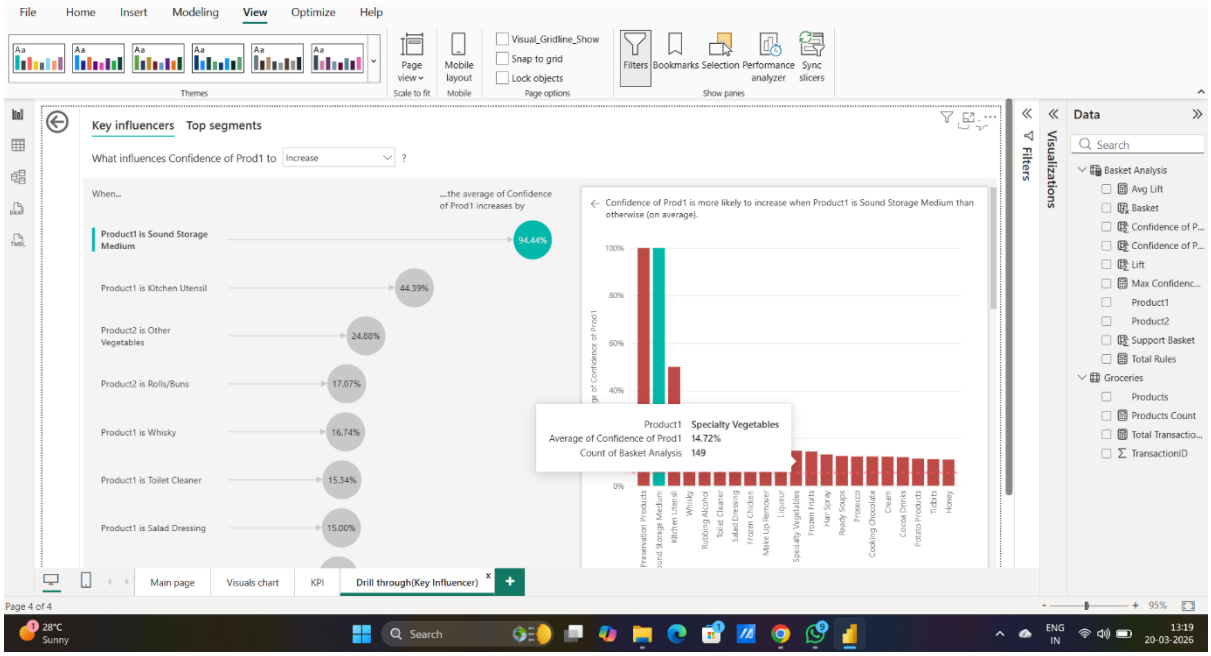
13:19
20-03-2026


File
Home
Insert
Modeling
View
Optimize
Help
Key influencers
Top segments
What influences Confidence of Prod1 to
Increase
?
When...
...the average of Confidence of Prod1 increases by
Product1 is Sound Storage Medium
94.44%
Product1 is Kitchen Utensil
44.39%
Product2 is Other Vegetables
24.88%
Product2 is Rolls/Buns
17.07%
Product1 is Whisky
16.74%
Product1 is Toilet Cleaner
15.34%
Product1 is Salad Dressing
15.00%
Confidence of Prod1 is more likely to increase when Product1 is Sound Storage Medium than otherwise (on average).
100%
80%
60%
40%
0%
Product1 Specialty Vegetables
Average of Confidence of Prod1 14.72%
Count of Basket Analysis 149
Preservation Products
Sound Storage Medium
Kitchen Utensil
Whisky
Rubbing Alcohol
Toilet Cleaner
Salad Dressing
Frozen Chicken
Make Up Remover
Liqueur
Specialty Vegetables
Frozen Fruits
Hair Spray
Ready Soups
Prosecco
Cooking Chocolate
Cream
Cocoa Drinks
Potato Products
Tidbits
Honey
Main page
Visuals chart
KPI
Drill through(Key Influencer)
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Data
Search
Basket Analysis
Avg Lift
Basket
Confidence of P...
Confidence of P...
Lift
Max Confidenc...
Product1
Product2
Support Basket
Total Rules
Groceries
Products
Products Count
Total Transactio...
TransactionID
Filters
Visualizations
28°C
Sunny
Search
ENG
IN
13:19
20-03-2026

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Code JupyterLab Python 3 (ipykernel)

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

[2]:
```
df=pd.read_csv("C:/Users/ASUS/OneDrive/Desktop/MARKET BASKET ANALYSIS/DATA/Groceries_dataset.csv")
```

[3]:
```
df.head()
```

[3]:

	Member_number	Date	itemDescription
0	1808	21-07-2015	tropical fruit
1	2552	05-01-2015	whole milk
2	2300	19-09-2015	pip fruit
3	1187	12-12-2015	other vegetables
4	3037	01-02-2015	whole milk

[4]:
```
df.columns
```

[4]:
```
Index(['Member_number', 'Date', 'itemDescription'], dtype='object')
```

[5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 38765 entries, 0 to 38764
Data columns (total 3 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Member_number    38765 non-null  int64
```

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Code JupyterLab Python 3 (ipykernel)

[6]:
```
df.describe()
```

[6]:

	Member_number
count	38765.000000
mean	3003.641868
std	1153.611031
min	1000.000000
25%	2002.000000
50%	3005.000000
75%	4007.000000
max	5000.000000

[7]:
```
df.isnull().sum().sort_values(ascending = False )
```

[7]:
```
Member_number      0
Date               0
itemDescription    0
dtype: int64
```

TYPE CONVERSION

[8]:
```
df["date"]=pd.to_datetime(df["Date"])
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 38765 entries, 0 to 38764
Data columns (total 4 columns):
 #   Column           Non-Null Count  Dtype
```

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Code JupyterLab Python 3 (ipykernel)

[9]:
```
df.head()
```

[9]:

	Member_number	Date	itemDescription	date
0	1808	21-07-2015	tropical fruit	2015-07-21
1	2552	05-01-2015	whole milk	2015-01-05
2	2300	19-09-2015	pip fruit	2015-09-19
3	1187	12-12-2015	other vegetables	2015-12-12
4	3037	01-02-2015	whole milk	2015-02-01

[10]:
```
item_distr = df.groupby(by = "itemDescription").size().reset_index(name="Frequency").sort_values(by ="Frequency",ascending=False).head(10)
bars = item_distr["itemDescription"]
height = item_distr["Frequency"]
x_pos = np.arange(len(bars))

plt.figure(figsize=(16,9))
plt.bar(x_pos,height,color=(0.2,0.3,0.5,0.5))
plt.title("Top 10 solid item ")
plt.xlabel("item names")
plt.ylabel("number of quantity sold")
plt.xticks(x_pos,bars)
plt.show()
```

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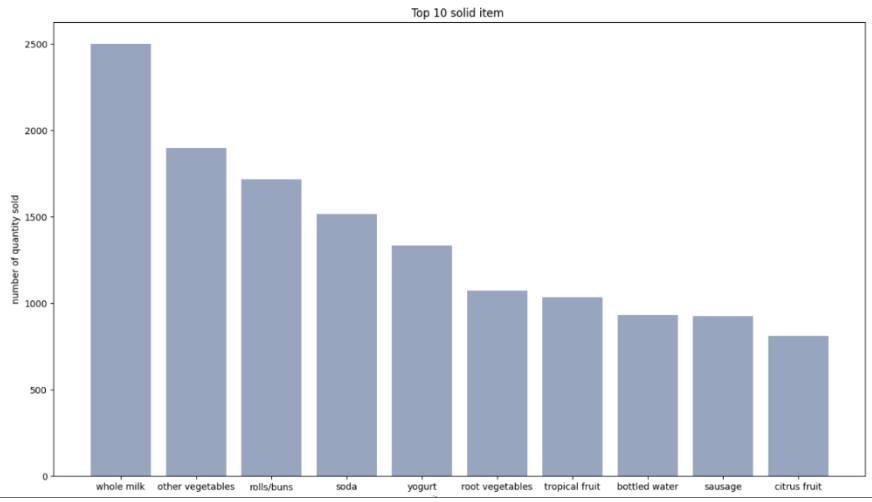
Code JupyterLab Python 3 (ipykernel)



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Code JupyterLab Python 3 (ipykernel)

[11]:
```
df_date=df.set_index(["Date"])
df_date
```

[11]:

	Member_number	itemDescription	date
Date			
21-07-2015	1808	tropical fruit	2015-07-21
05-01-2015	2552	whole milk	2015-01-05
19-09-2015	2300	pip fruit	2015-09-19
12-12-2015	1187	other vegetables	2015-12-12
01-02-2015	3037	whole milk	2015-02-01
...	...	...	...
08-10-2014	4471	sliced cheese	2014-10-08
23-02-2014	2022	candy	2014-02-23
16-04-2014	1097	cake bar	2014-04-16
03-12-2014	1510	fruit/vegetable juice	2014-12-03
26-12-2014	1521	cat food	2014-12-26

38765 rows × 3 columns

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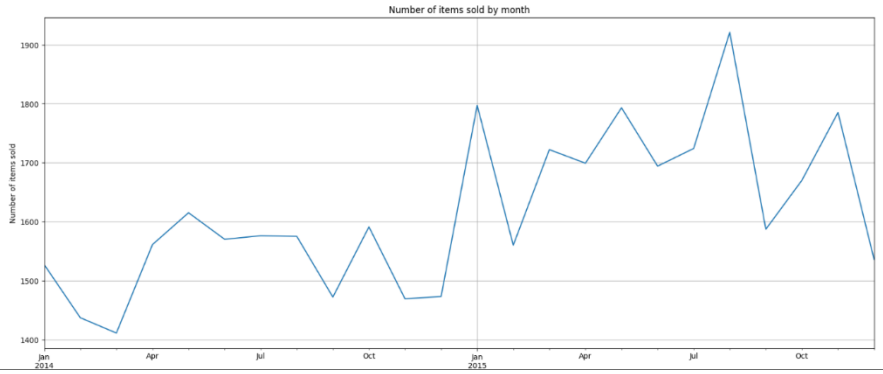
Code JupyterLab Python 3 (ipykernel)

[12]:
```
df_date['date'] = pd.to_datetime(df_date['date'])

df_date.resample("ME", on='date')['itemDescription'].count().plot(
    figsize=(20,8),
    grid=True,
    title="Number of items sold by month"
).set(xlabel="Date", ylabel="Number of items sold")
```

[12]: [Text(0.5, 0, 'Date'), Text(0, 0.5, 'Number of items sold')]



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Code JupyterLab Python 3 (ipykernel)

[13]:
```
cust_level=df[["Member_number","itemDescription"]].sort_values(by = "Member_number",ascending=False)
cust_level["itemDescription"]=cust_level["itemDescription"].str.strip()
cust_level
```

[13]:

	Member_number	itemDescription
34885	5000	semi-finished bread
25489	5000	other vegetables
3578	5000	soda
27877	5000	onions
9340	5000	bottled beer
...	...	...
24544	1000	yogurt
4843	1000	sausage
20992	1000	semi-finished bread
8395	1000	whole milk
13331	1000	whole milk

38765 rows × 2 columns

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Code JupyterLab Python 3 (ipykernel)

[14]:
```
transactions = [a[1]["itemDescription"].tolist() for a in list(cust_level.groupby(["Member_number"]))]
```

[15]:
```
pip install apyori
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: apyori in .\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.13_qbz5n2kfra8p0\LocalCache\local-packages\Python31
3\site-packages (1.1.2)
Note: you may need to restart the kernel to use updated packages.
```

[ ]:

[ ]:

[16]:
```
from apyori import apriori
```

[17]:
```
rules = apriori(
    transactions=transactions,
    min_support=0.005,   # lower support
    min_confidence=0.2,
    min_lift=1,
    min_length=2
)
```

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Code JupyterLab Python 3 (ipykernel)

[18]:
```
results=list(rules)
```

[19]:
```
results
```

[19]:
```
[RelationRecord(items=frozenset({'bottled water'}), support=0.21369933299127758, ordered_statistics=[OrderedStatistic(items_base=frozenset(), items_a
dd=frozenset({'bottled water'}), confidence=0.21369933299127758, lift=1.0)]),
 RelationRecord(items=frozenset({'other vegetables'}), support=0.3766033863519754, ordered_statistics=[OrderedStatistic(items_base=frozenset(), items
_add=frozenset({'other vegetables'}), confidence=0.3766033863519754, lift=1.0)]),
 RelationRecord(items=frozenset({'rolls/buns'}), support=0.3496664956387891, ordered_statistics=[OrderedStatistic(items_base=frozenset(), items_add=f
rozenset({'rolls/buns'}), confidence=0.3496664956387891, lift=1.0)]),
 RelationRecord(items=frozenset({'root vegetables'}), support=0.23063109286813752, ordered_statistics=[OrderedStatistic(items_base=frozenset(), items
_add=frozenset({'root vegetables'}), confidence=0.23063109286813752, lift=1.0)]),
 RelationRecord(items=frozenset({'sausage'}), support=0.2060030785017958, ordered_statistics=[OrderedStatistic(items_base=frozenset(), items_add=froz
enset({'sausage'}), confidence=0.2060030785017958, lift=1.0)]),
 RelationRecord(items=frozenset({'soda'}), support=0.3134940995382247, ordered_statistics=[OrderedStatistic(items_base=frozenset(), items_add=frozens
et({'soda'}), confidence=0.3134940995382247, lift=1.0)]),
 RelationRecord(items=frozenset({'tropical fruit'}), support=0.2337095946639302, ordered_statistics=[OrderedStatistic(items_base=frozenset(), items_a
dd=frozenset({'tropical fruit'}), confidence=0.2337095946639302, lift=1.0)]),
 RelationRecord(items=frozenset({'whole milk'}), support=0.4581836839404823, ordered_statistics=[OrderedStatistic(items_base=frozenset(), items_add=f
rozenset({'whole milk'}), confidence=0.4581836839404823, lift=1.0)]),
 RelationRecord(items=frozenset({'yogurt'}), support=0.28296562339661363, ordered_statistics=[OrderedStatistic(items_base=frozenset(), items_add=froz
enset({'yogurt'}), confidence=0.28296562339661363, lift=1.0)])
```

jupyter MBA Last Checkpoint: yesterday

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Code JupyterLab Python 3 (ipykernel)

[36]:
```
def inspect(results):
    data = []

    for result in results:
        support = result[1]

        for rule in result[2]:
            lhs = list(rule[0])
            rhs = list(rule[1])
            confidence = rule[2]
            lift = rule[3]

            data.append((lhs, rhs, support, confidence, lift))

    return data


resultsindataframe = pd.DataFrame(
    inspect(results),
    columns=['Left Hand Side', 'Right Hand Side', 'Support', 'Confidence', 'Lift']
)

print(resultsindataframe.head())
```

```
  Left Hand Side     Right Hand Side   Support  Confidence  Lift
0             []     [bottled water]  0.213699    0.213699   1.0
1             []  [other vegetables]  0.376603    0.376603   1.0
2             []        [rolls/buns]  0.349666    0.349666   1.0
3             []   [root vegetables]  0.230631    0.230631   1.0
4             []           [sausage]  0.206003    0.206003   1.0
```

[ ]:

[ ]:

5. Limitations

Despite its usefulness, Market Basket Analysis has certain limitations:

- **Dependence on Data Quality:**
  The accuracy of results depends on the quality and completeness of the dataset. Missing or noisy data can lead to incorrect associations.
- **Ignores Time-Based Behavior:**
  The analysis does not consider when purchases occur, so seasonal or time-based patterns are not captured.
- **Large Dataset Complexity:**
  As the dataset size increases, the number of possible item combinations grows rapidly, making computation more complex and time-consuming.
- **Limited Context Awareness:**
  The model only considers item co-occurrence and does not account for external factors such as customer preferences, price, or promotions.
- **May Generate Irrelevant Rules:**
  Some association rules may not be meaningful or useful for business decisions, especially if thresholds are not properly set.
- **Static Analysis:**
  The results are based on historical data and may not reflect real-time changes in customer behavior.
- **Interpretation Challenges:**
  Understanding and selecting the most useful rules (based on Support, Confidence, and Lift) can be difficult without proper visualization tools like Microsoft Power BI.

6. Conclusion

The Market Basket Analysis project successfully demonstrates how data mining techniques can be used to uncover hidden patterns and relationships in transactional data. By applying the Apriori algorithm, meaningful association rules were generated, helping to identify products that are frequently purchased together.

The use of key metrics such as Support, Confidence, and Lift enabled effective evaluation of the strength and reliability of these associations. The integration of these results into an interactive dashboard using Microsoft Power BI made the analysis more intuitive and user-friendly.

Various visualizations such as scatter plots, bar charts, doughnut charts, and key influencer visuals provided deeper insights into customer purchasing behavior. Additionally, the drill-through functionality enhanced the interactivity of the dashboard by allowing detailed product-level analysis.

Overall, this project highlights the importance of combining data mining techniques with visualization tools to support data-driven decision-making. It can be effectively used in retail and e-commerce industries to improve product placement, recommendation systems, and marketing strategies.